



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

EE2104-1 – GENERATION TRANSMISSION AND DISTRIBUTION

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Demand factor is defined as**
A) Average load/ Maximum load
B) Connected load/ Maximum demand
C) Maximum demand / Connected load
D) Average load × Maximum load
2. **Which of the following is the essential requirement of peak load plant**
A) It must run at high speed
B) It must produce high voltage
C) It must be small in size
D) It must be capable of starting quickly
3. **The main objective of tariff is to distribute equally the cost of**
A) Energy production among all its consumers
B) Power transmission and distribution
C) Supplying energy among the various classification of users
D) Installation and fuel
4. **A diesel power plant is best suited as**
A) Base load plant
B) Stand-by plant
C) Peak load plant
D) General purpose plant
5. **In a superheater**
A) Pressure rises and temperature drops
B) Temperature rises and Pressure remains unchanged
C) Temperature rises and Pressure drops
D) Pressure rises and temperature remains the same
6. **Which of the following generating station has minimum running cost**
A) Nuclear
B) Thermal
C) Hydro
D) Natural Gas
7. **In a load curve, the highest point represents**
A) Peak demand
B) Average demand
C) Diversified demand
D) Minimum demand
8. **What is the effect of rise in temperature on sag when ice and wind effect are eliminated**
A) Sag decreases
B) Sag increases
C) Sag remains constant
D) Sag becomes zero
9. **The corona discharge on transmission lines may be avoided by**
A) Increasing effective conductor radius
B) Increasing the operating voltage
C) Decreasing the spacing between the conductors
D) Increasing the power
10. **When does the Ferranti effect occur on the transmission line**
A) When the line is short and loaded
B) When the line is long and unloaded
C) When the line is long and loaded
D) When the line is short and unloaded
11. **What are the values of A, B, C, D parameters of a short transmission line?**
A) Z, 0, 1, 1
B) 1, Z, 0, 1
C) 0, 1, 1, 1
D) 1, 1, Z, 0
12. **Bundled conductors in EHV transmission system provide**
A) Reduced capacitance
B) Increased inductance
C) Increased capacitance
D) Increased voltage gradient

13. **Transmission efficiency of a transmission line increases with the**
 A) Decrease in power factor and voltage B) Increase in power factor and voltage
 C) Increase in power factor but decrease in voltage D) Increase in voltage and decrease in power factor
14. **Transposition of transmission line is done to**
 A) Reduce corona B) Balance line voltage drop
 C) Reduced skin effect D) Reduce line loss
15. **What is meant by capacitance grading of cables**
 A) Use of dielectrics in different concentrations B) Introduction of capacitance at various lengths of cable to counter the effect of inductance
 C) Use of dielectrics of different permittivities D) Grading according to capacitance per km length of the cable
16. **The rated voltage of a 3 phase power system is given by**
 A) rms peak voltage B) Peak phase voltage
 C) rms line to line voltage D) Peak line to line voltage
17. **FACTS devices used in**
 A) Generation B) AC transmission
 C) DC transmission D) Small power system
18. **PVC stands for**
 A) Polyvinyl chloride B) Post varnish conductor
 C) Pressed and varnished cloth D) Positive voltage conductor
19. **Underground cables are laid at sufficient depth**
 A) To minimise temperature stresses B) To avoid being unearthed easily due to removal of soil
 C) To minimise the effect of shocks and vibrations due to passing vehicles, etc. D) To increase the temperature stress
20. **The insulation of the cable decreases with**
 A) The increase in length of the insulation B) The decrease in the length of the insulation
 C) Decreased temperature D) Decreased humidity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|----------|
| 1. a) With a typical layout explain the functioning of hydroelectric power plant. | 8 | L*2 | 1 | 1,2,9,10 |
| b) Define the following: -
i. Load duration curve
ii. Demand factor
iii. Plant use factor
iv. Maximum demand | 4 | L1 | 2 | 1,2,9,10 |
| c) Explain the categories of cost of electrical energy. | 4 | L2 | 2 | 1,2,9,10 |
| 2. a) Explain the working of thermal power plant with a neat schematic diagram. | 8 | L2 | 1 | 1,2,9,10 |
| b) The capital cost of a hydro-power station of 100 MW capacity is Rs. 1000 per kW. The annual depreciation charges are 15 % of the capital cost. A royalty of Rs. 2 per kW per year and Rs. 0.03 per kWh generated is to be paid for using the river water for generation of power. The maximum demand on the power station is 70 MW and annual load factor is 60 %. Annual cost of salaries, maintenance charges etc. is Rs. 1000000. If 20 % of this expense is also chargeable as fixed charges, calculate the generation cost in two-part form. | 8 | L3 | 2 | 1,2,9,10 |
| 3. a) With a neat diagram, illustrate the functioning of diesel power plant. | 8 | L2 | 1 | 1,2,9,10 |

- b) Plot the hydrograph and find the average discharge available for the whole period from the following average weekly discharge measured at a site.

Week	Discharge Q (m ³ / sec)
1	200
2	300
3	1100
4	700
5	900
6	800
7	600
8	1000
9	500
10	400
11	500
12	700
13	100

- c) List the Pros and Cons of nuclear power generation.

5 L3 2 1,2,9,10
3 L1 1 1,2,9,10

Unit – II

4. a) Obtain an expression for sag of a transmission line when suspended on an unequal level supports.
b) Explain Ferranti effects in transmission lines.
c) Explain the methods to reduce corona effect in transmission line.
5. a) Obtain an expression for sending end voltage and current of a short transmission line.
b) Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is $\frac{1}{8}$ th of the capacitance of the insulator itself. Also find the string efficiency.
6. a) What are the various types of insulators? With the neat figures explain any two of them.
b) A single-phase transmission line delivers 2 MW of power at the receiving end at a voltage of 33 kV and 0.9 power factor lagging. The total resistance of the line is 10 Ω and the total inductive reactance is 18 Ω . Determine:
i. Percentage voltage regulation
ii. Sending end power factor
iii. Transmission efficiency

8 L3 3 1,2,9,10
4 L2 4 1,2,9,10
4 L2 4 1,2,9,10
8 L3 4 1,2,9,10
8 L3 3 1,2,9,10
8 L2 3 1,2,9,10
8 L3 4 1,2,9,10

Unit – III

7. a) Obtain an expression for dielectric stress in a single core cable.
b) Calculate the capacitance and charging current of a single-core cable used on a 3-phase 66 kV system. The cable is 1 km long having a core diameter of 15 cm and sheath diameter 60 cm. The relative permittivity of the insulation may be taken as 3.6 and the supply frequency 50 Hz.
c) Explain any three commonly used insulating materials in underground cables.
8. a) Obtain an expression for sending voltage considering power factor with respect to receiving end of the distributor.
b) State the advantages and disadvantages of HVDC transmission.

8 L3 5 1,2,9,10
5 L3 5 1,2,9,10
3 L2 5 1,2,9,10
8 L3 5 1,2,9,10
8 L1 5 1,2,9,10

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
